

Drug Runner

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Summary

Two listening posts pick up the same radio signal from a point located somewhere in the ocean. This signal is presumed to come from a boat waiting to deliver drugs. A helicopter is dispatched to the scene, but how does it find the radio source without radar?

Two different models give varying search patterns based on different assumptions about how the boat might move. The first assumes that the boat will move perpendicularly to the path of the helicopter and that the boat does not leave a wake. This model represents the places where the boat might be as vectors and follows them over time. The helicopter always moves towards the region of greatest probability.

The second model assumes that the boat moves away from the first sound of the helicopter and does leave a wake. The helicopter covers a zigzag path across the region. The drawback to this solution is that the helicopter cannot cover the entire region before it has to return to base. However, the helicopter still has a 91% chance of finding the boat.

Various other models were considered, and their advantages and disadvantages are summarized. These include different escape methods, and various search paths. One of the primary factors is the presence or absence of a wake behind the boat. If the weather is calm, then the wake should be easy to see; but in even a light wind, the wake will be obscured by other waves.

The second solution, with a 91% chance of finding the boat, seems to be the best. However, installing radar or increasing the range of the helicopter could bring the chance of finding the boat much closer to 100%.

"One of the primary factors is the presence or absence of a wake."

Assumptions

1. The probability distribution of the angles is a normal distribution.
2. The helicopter is similar to a OH-58 Kiowa, a small military helicopter.
 - a. There is an instantaneous view from the spotlight over a full 200-ft-radius semicircle. The searchlight moves fast enough to cover effectively the whole semicircle.
 - b. The helicopter has a speed of 140.4 miles per hour and a range of 287.5 miles.
 - c. The helicopter will not return to base and refuel, because the drugs would be transferred by the time the helicopter could return.
 - d. The helicopter flies at a constant height.
3. The people on the boat hear the helicopter before they see the helicopter or the spotlight.
4. It is impossible for the people in the helicopter to see the boat if it is outside the spotlight.
5. The boat may escape in one of three patterns:
 - a. A random direction
 - b. In a direction exactly opposite to the direction from where the boat first hears the helicopter
 - c. In a direction perpendicular to the path of the helicopter.
6. The boat may or may not produce a wake. If it does:
 - a. The boat does not plane.
 - b. Three wakes can be detected (0° , 19.5° , -19.5°).
 - c. Wakes exist for an infinite amount of time.
 - d. Wakes do not spread behind the initial position of the boat.
 - e. The wake advances at a constant velocity.

Probability Functions

The problem states that the radar detectors used to locate the boat originally are accurate to within 2° . This implies that there is a some sort of a probability distribution of the possible angles at each station. We assumed that this is a normal probability distribution.

The normal distribution with mean μ and standard deviation σ has probability density function

$$f(x) = \exp\left\{-\frac{1}{2}\left[(x - \mu)/\sigma\right]^2\right\} / \sqrt{2\pi}, \quad -\infty < x < \infty$$

We interpret the measurement accuracies in the problem statement to be 95% confidence intervals and hence to encompass two

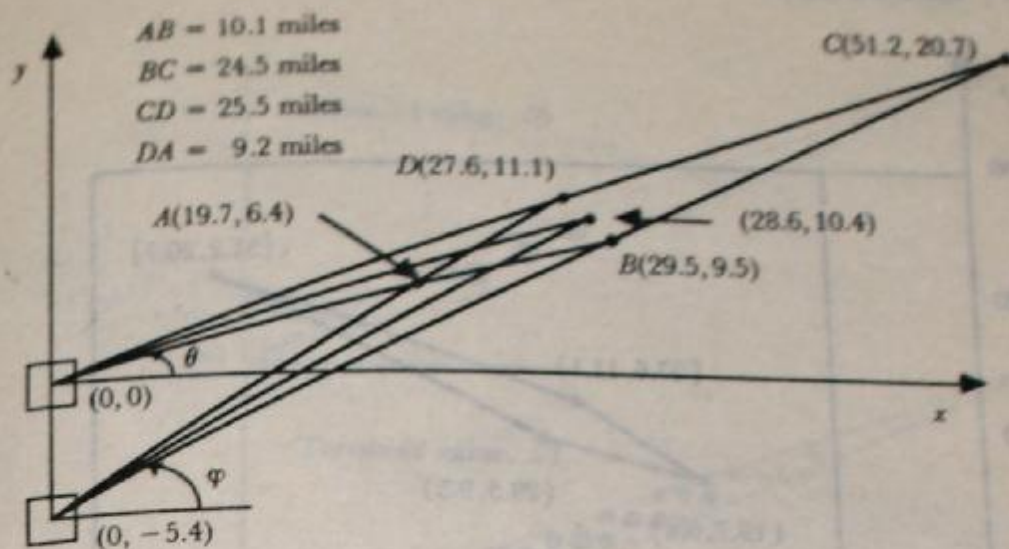


Figure 1. Geometry of the problem. Listening posts are at (0, 0) and (0, -5.4).

standard deviations to each side of the mean, so that the standard deviation for each angle measurement is 1° .

We assign origin and coordinate axes as shown in Figure 1, and we consequently measure angles counterclockwise relative to the positive x-axis.

We have two different probability functions, one for each radar station and its angle (which we measure in degrees):

$$f_\theta(\theta) = \exp\left[-\frac{1}{2}(\theta - 20^\circ)^2\right] / \sqrt{2\pi}, \quad -\infty < \theta < \infty,$$

$$f_\varphi(\varphi) = \exp\left[-\frac{1}{2}(\varphi - 29^\circ)^2\right] / \sqrt{2\pi}, \quad -\infty < \varphi < \infty.$$

The two angles are independent random variables, so their joint density is

$$g(\theta, \varphi) = \exp\left[-\frac{1}{2}(\theta - 20^\circ)^2 - \frac{1}{2}(\varphi - 29^\circ)^2\right] / 2\pi, \\ -\infty < \theta < \infty, \quad -\infty < \varphi < \infty.$$

In fact, however, we do not wish to consider the angles over such broad domains. Instead, we transform to (x, y) coordinates, truncate the domain to $10 < x < 60$ and $0 < y < 30$ (see Figure 2) and renormalize (so that total probability over the new domain is 1).

In Figures 3 and 4 a crude grid is laid over the domain of the joint density function and densities are shown at the grid points.

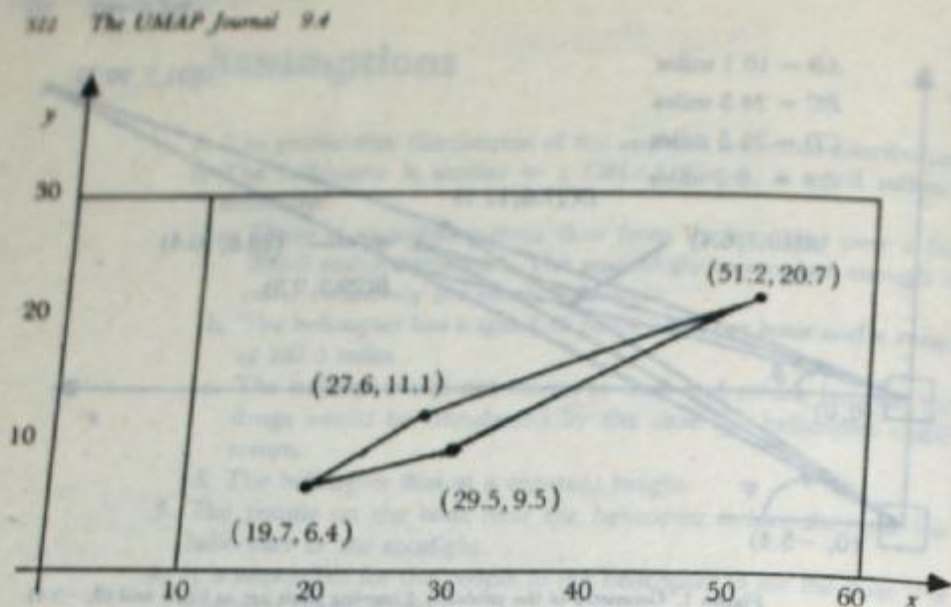


Figure 2. Restricted domain for the final density function.

Figure 3 displays densities at different thresholds. Figure 4 exhibits a crude approximation to the probability surface.¹

Does the Boat Escape?

We considered the following escape methods for the boat (see Figure 5):

- A. In a random direction
- B. In a direction exactly opposite to the direction from which the boat first hears the helicopter
- C. In a direction perpendicular to the path of the helicopter

We considered two different models when dealing with boat escape. The first assumes that the wake cannot be detected, the second considers the wake produced by the boat.

In the first case, the boat may try to escape in any of the above three fashions (A, B, or C).

¹EDITOR'S NOTE. The edited formulation above interprets and elaborates the authors' intended meaning. In actual fact, they expressed the joint density function directly in (x, y) coordinates but did not incorporate the Jacobian of the transformation $(\theta, \varphi) \rightarrow (x, y)$ in the renormalization calculation. This oversight affects the probabilities they calculate but does not invalidate their approach.

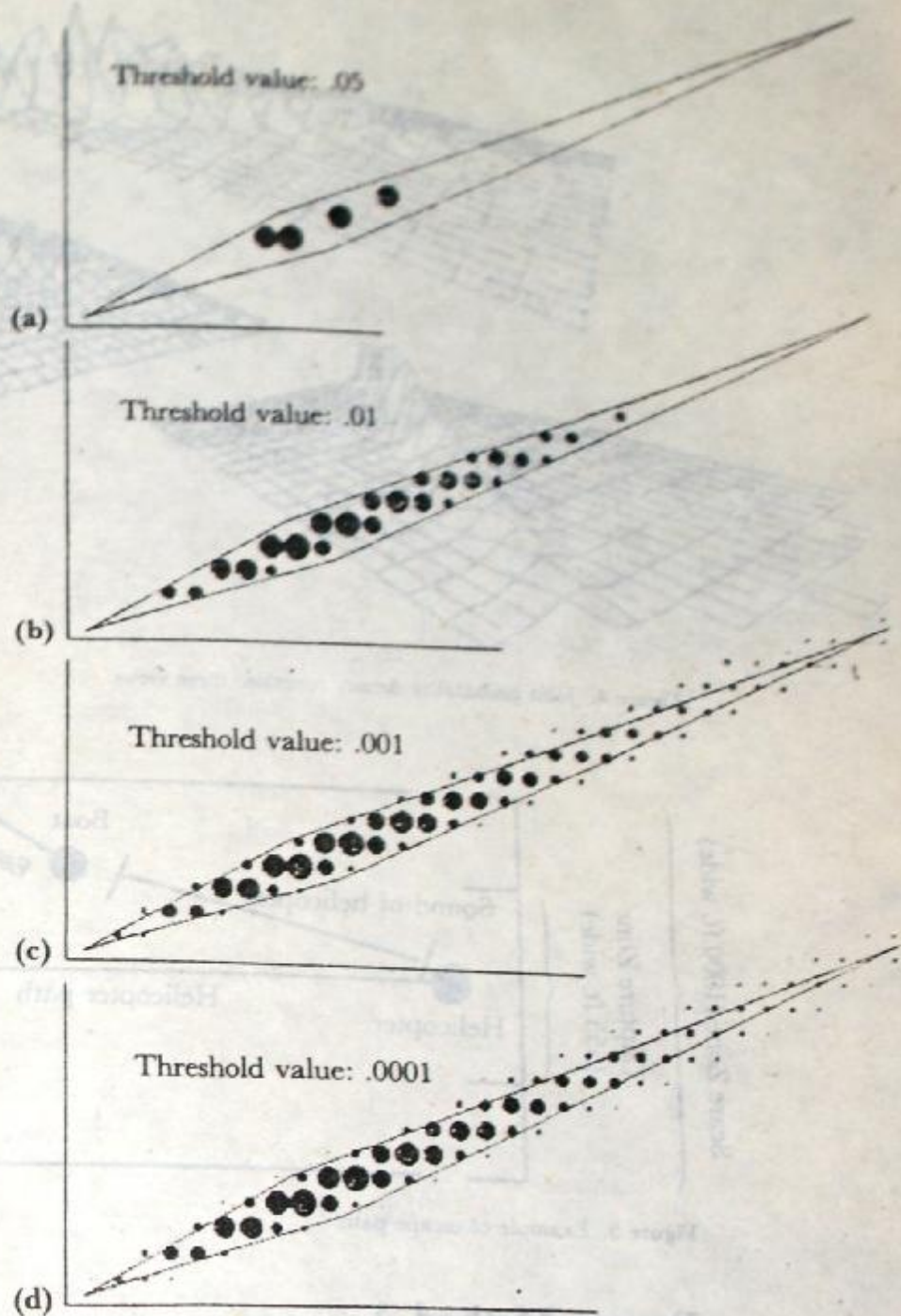


Figure 3. Joint probability density function at grid points. Threshold is the minimum probability displayed. The radius of a circle is proportional to the square root of the probability at the point.

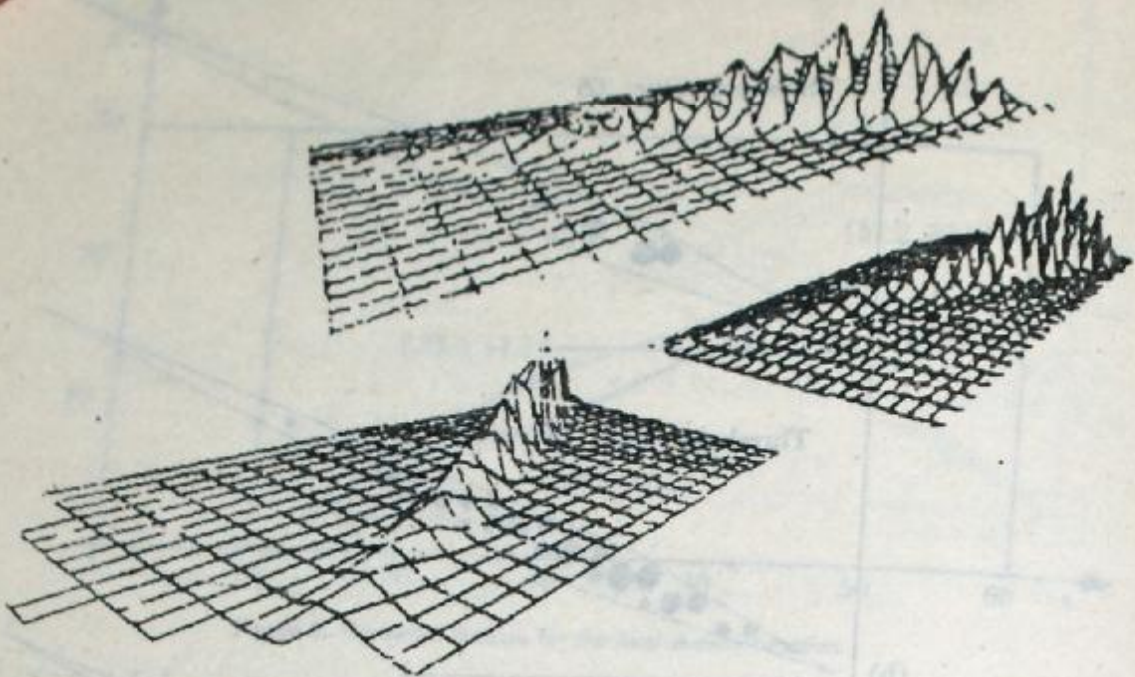


Figure 4. Joint probability density function: three views.

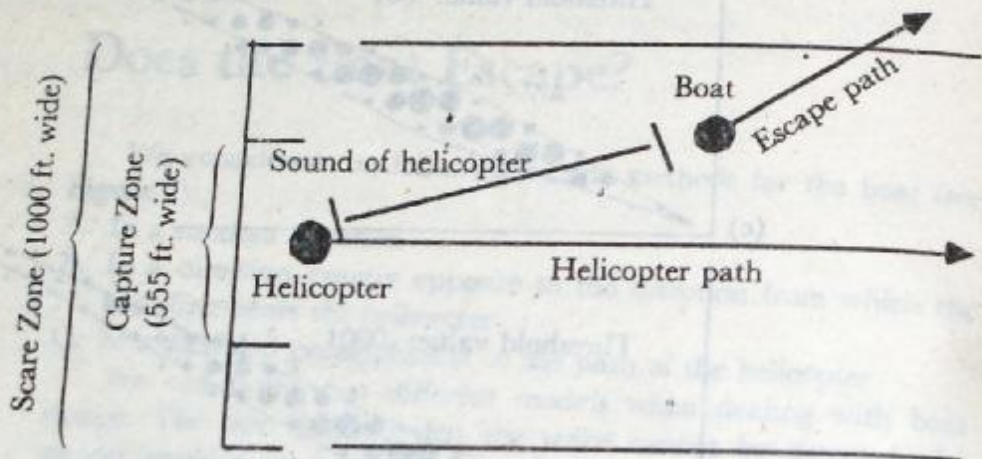


Figure 5. Example of escape path.

Escape Method A

The first escape method (A) is to move in any random direction. This would only happen if it is hard to identify the direction from which the helicopter is coming or if the drug dealers panic. However, on a perfectly still night, it should be easy to identify the sound. If the drug dealers panic, then they aren't going to last in the business

long. Anyway, if the boat moves randomly there is a chance that it is able to escape. Even if it is very near the searchlight of the boat, it can escape if it happens to move perpendicular to the helicopter's line of motion. It is more likely, though, that the boat ends up going in a direction towards the helicopter.

Motion is rarely totally random, however. The motion will most likely be distributed around the direction opposite to that from which the helicopter was heard. So the problem is not that different from assuming the boat moves directly away from the first sound of the helicopter. This is the second escape method (B).

Escape Method B

The helicopter spotlight has a maximum range of 225 ft, so the helicopter can search the area within 225 ft, on both sides of its flight path. If the boat leaves this corridor before the helicopter comes abreast of it the boat has escaped detection. Figure 6 illustrates the critical case, where the boat is at the edge of the corridor as the helicopter pulls abreast. The boat has gone a distance x at an angle to the helicopter path, so the helicopter has gone a distance $3x$.

By the Pythagorean theorem:

$$(x + 500)^2 = 225^2 + (3x)^2, \text{ so}$$

$$8x^2 - 1000x - 199375 = 0, \text{ and}$$

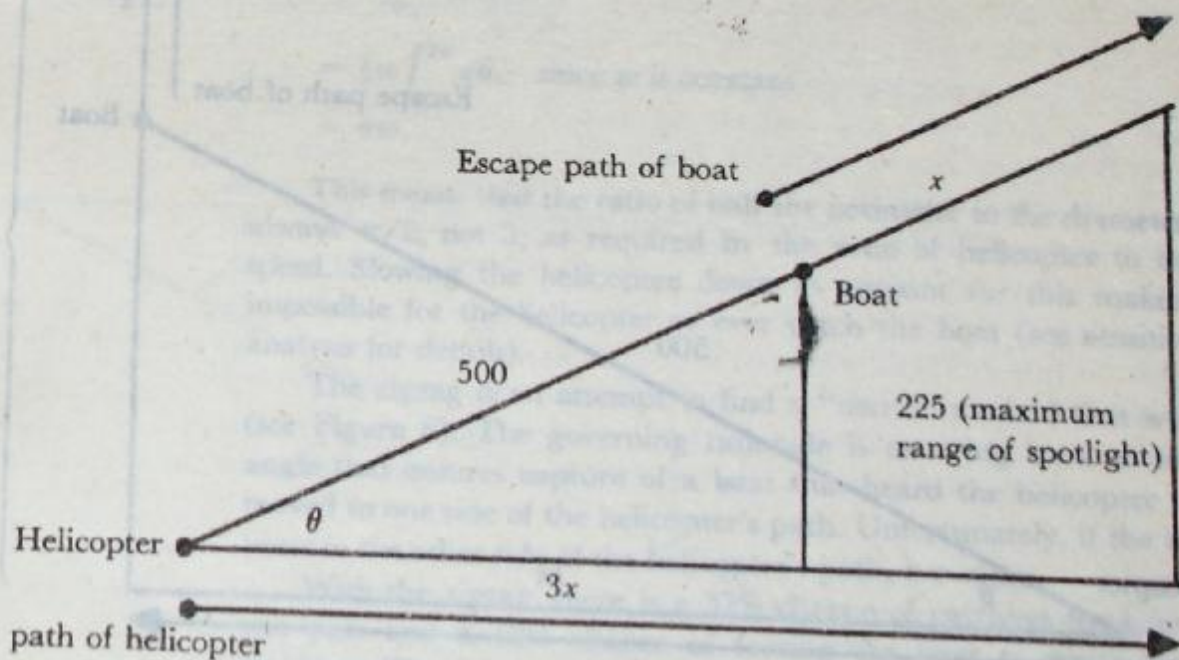


Figure 6. Finding escape distance for case of no boat wake and escape method B.

$$x = 232,$$

$$\theta = 17.9^\circ,$$

$$\text{escape distance} = 500 \sin \theta = 153.6 \text{ ft.}$$

So, for escape method B, the boat will be captured only if it is within 153.6 ft of the path of the helicopter. Therefore, the helicopter captures the boat if it is in a path 307 ft wide and forces the boat to move (leaving no wake to follow) if it is in the other 693 ft of width where the helicopter can be heard.

Escape Method C

The third escape method (C) is escape perpendicular to the path of the helicopter. For the analysis we generalize and let k be the ratio of helicopter speed to boat speed. Figure 7 illustrates the critical case.

By the Pythagorean theorem:

$$500^2 = (225 - x)^2 + (kx)^2, \text{ so}$$

$$(k^2 + 1)x^2 - 450x - 199375 = 0, \text{ and}$$

$$x = \frac{450 + \sqrt{1000000 + 797500k^2}}{2(k^2 + 1)}.$$

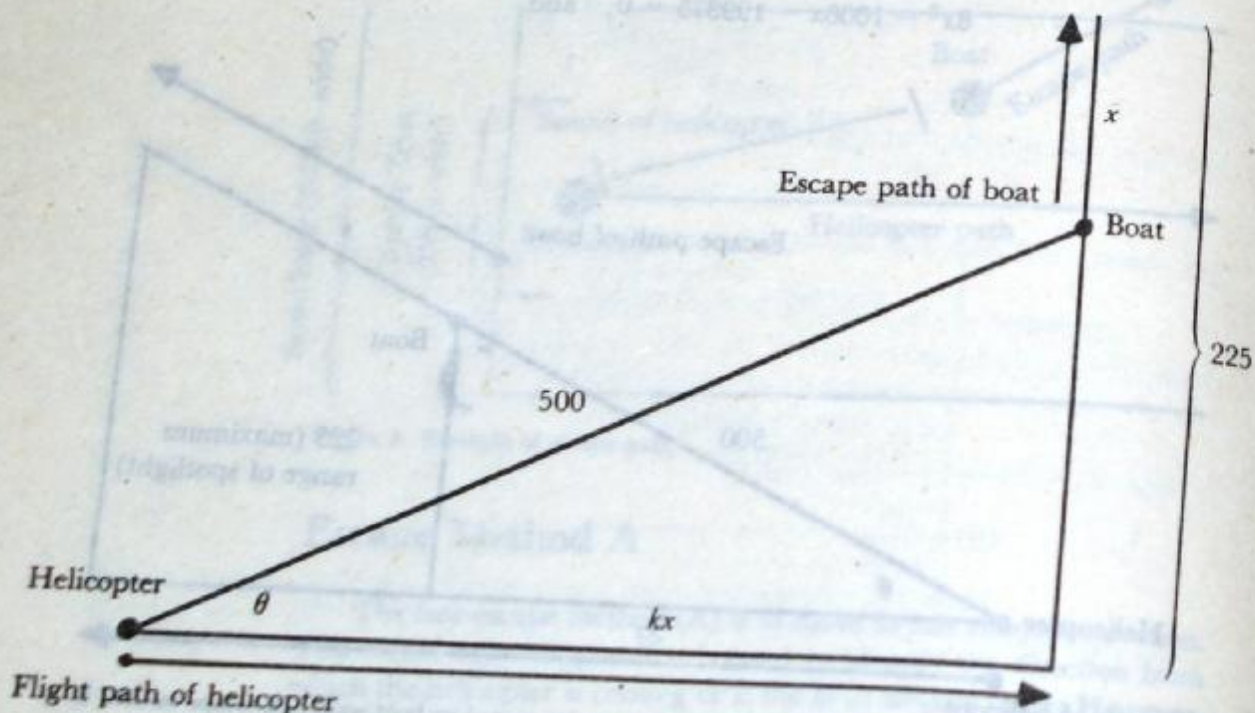


Figure 7. Finding escape distance for the case of no boat wake and escape method C.

If $k = 3$, then:

$$x = 165.5,$$

$$\theta = 6.8^\circ,$$

$$\text{escape distance} = 500 \sin \theta = 59.5 \text{ ft.}$$

So, for escape method C, the boat will be captured only if it is within 59.5 ft of the path of the helicopter. Therefore, the helicopter captures the boat if it is in path 119 ft wide, and forces the boat to move if it is in the other 881 ft, where the helicopter can be heard. Two types of searches were studied for the perpendicular case: the "corral" method, and the "most probable region" method.

Escape Method C: "Corral" Search Method

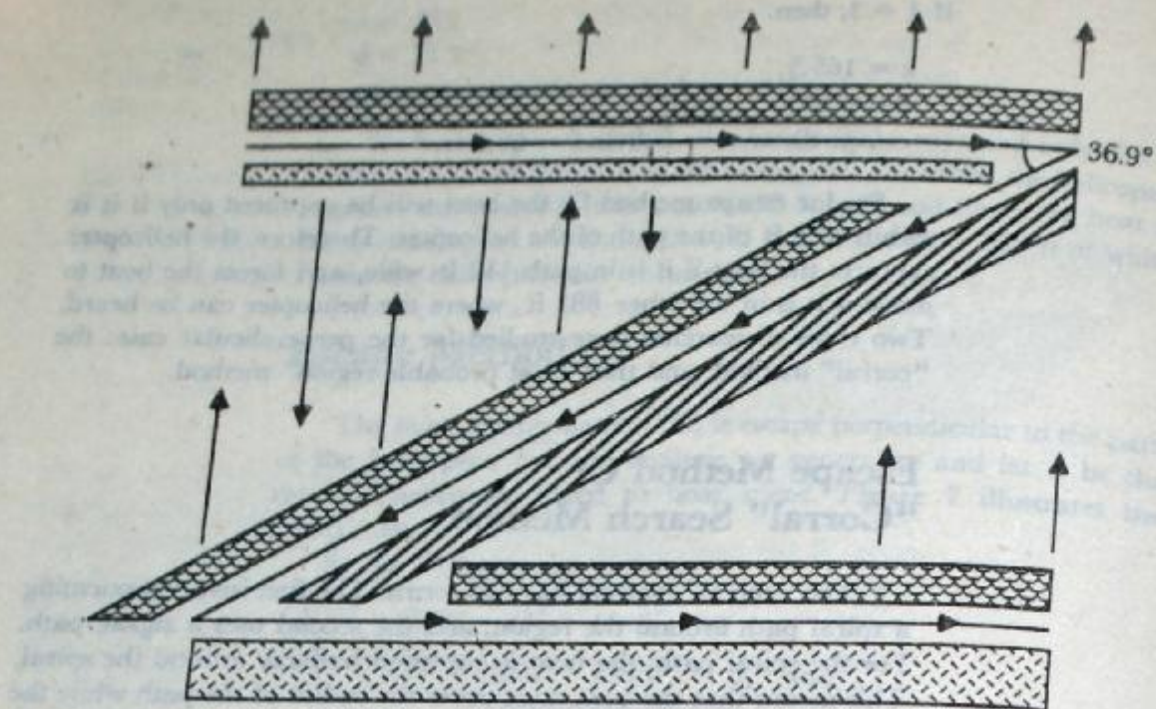
The "corral" method has two forms. The first involves pursuing a spiral path around the region, and the second uses a zigzag path. For the spiral path, the boat is captured halfway around the spiral. This means that the boat must cover the radius of the path while the helicopter, at three times the speed, must cover one-half of its perimeter. For this to hold at any point on the path, the path must be a curve of constant width. From a theorem of Cauchy, the perimeter of a curve of width w is

$$\begin{aligned} P &= \frac{1}{2} \int_0^{2\pi} w(\theta) d\theta \\ &= \frac{1}{2} w \int_0^{2\pi} d\theta, \text{ since } w \text{ is constant} \\ &= \pi w. \end{aligned}$$

This means that the ratio of half the perimeter to the diameter is always $\pi/2$, not 3, as required by the ratio of helicopter to boat speed. Slowing the helicopter down to account for this makes it impossible for the helicopter to ever catch the boat (see sensitivity analysis for details).

The zigzag is an attempt to find a "corral" method that works (see Figure 8). The governing rationale is to swing back at some angle that ensures capture of a boat that heard the helicopter and moved to one side of the helicopter's path. Unfortunately, if the boat veers to the other side of the helicopter's path, it escapes.

With the zigzag, there is a 37% chance of catching the boat in one pass and a 12% chance of forcing the boat to move but it escaping. There is a 51% chance that the boat does not move and could be dealt with later. These percentages assume equal probab-



Note: Arrows are scared boats.

Key: — Path of helicopter



Get away



Caught on diagonal run



Caught on next horizontal run

Figure 8. Zigzag with no wake.

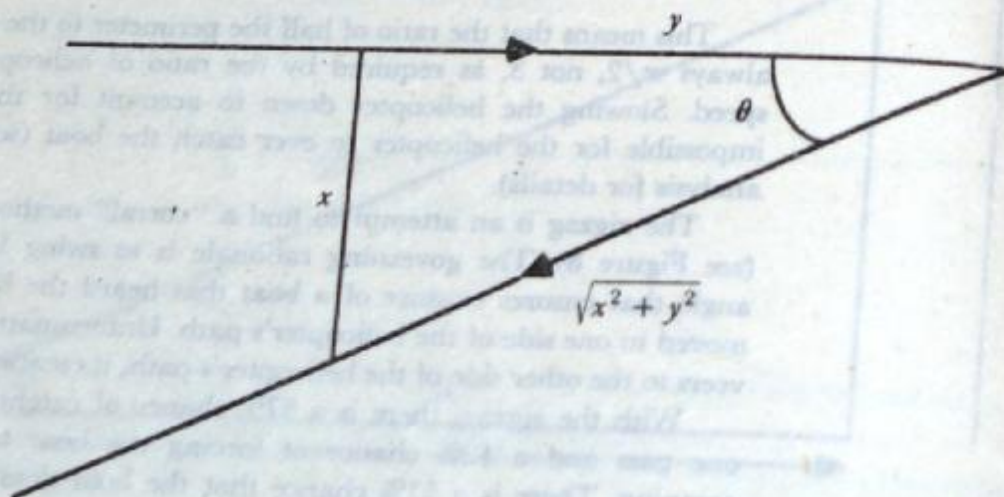


Figure 9. The zigzag critical case.

ity density for the boat at all points. This is not true, of course; but because of the low chance of catching the boat, the zigzag was abandoned before the true probability distribution was considered.

The critical case is shown in Figure 9.² While the boat travels a distance x , the helicopter travels a distance

$$\sqrt{x^2 + y^2} + y = 3x.$$

Solving

$$\sqrt{x^2 + y^2} = 3x - y,$$

$$x^2 + y^2 = 9x^2 - 6xy + y^2,$$

$$x^2 = 9x^2 - 6xy,$$

$$6xy = 8x^2,$$

$$y = 8x/6 = 4x/3, \text{ so}$$

$$\theta = \tan^{-1}(x/y) = \tan^{-1}(3/4) = 36.9^\circ.$$

Escape Method C: "Most Probable Region" Search Method

The "most probable region" method interprets the boat as a vector, moving away from the path of the helicopter. The helicopter then flies towards the region where the chance of finding a stationary or moving boat is the highest. This dynamic strategy was run as a computer simulation. The search domain was discretized to a grid with interval 333 ft, and the boat and helicopter are assumed to only move north-south and east-west from one gridpoint to another. This approach is not a feasible solution, as the full simulation took more than 3 hours, and the searchers could not wait that long for their simulation to run. A feasible search method must be easy to define, no matter what the angles read off the radar displays may be. Also the winding pattern would be hard to follow even if a Cray supercomputer were available to calculate the path (an unlikely prospect, since radar appears to be too much for the DEA's budget).

²EDITOR'S NOTE. The authors here regard the width of the capture corridor as negligible in comparison with the size of x and y .

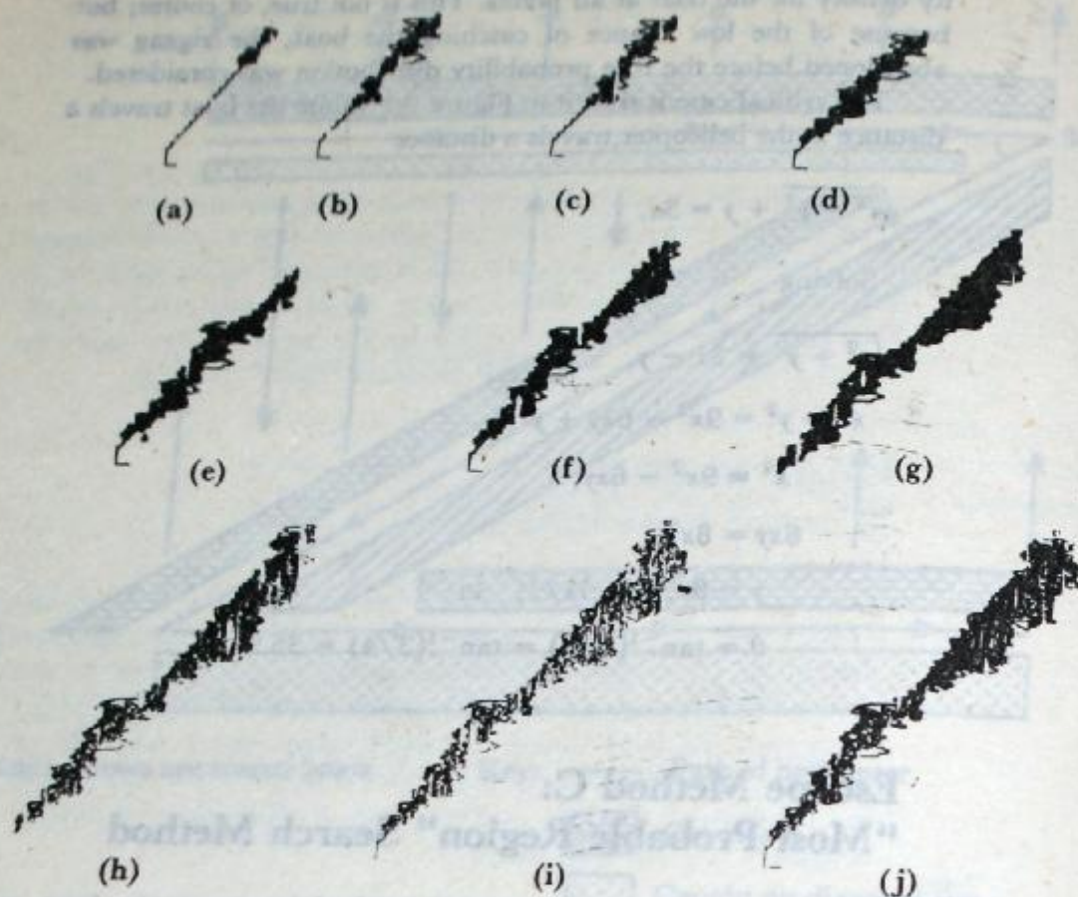


Figure 10. Output from the computer program simulating a search in the perpendicular escape (method B) and no wake case. The pictures are labeled in order of increasing time.

The Boat with Wake

Because the problem states that the weather is calm, it is reasonable to assume that the boat's wake is easy to see and lasts for a long time. The addition of the wake makes the searchers' job much easier. Once the boat hears the helicopter and moves, the observers in the helicopter need only spot the expanding wake, instead of a tiny boat. As Figure 11 illustrates, capture is ensured within the range of the problem.

The major problem with this approach is that the enormous downdraft of the helicopter creates a wake. However, this helicopter wake is quite small compared to the wake of the boat. Since the helicopter wake spreads behind the helicopter (the velocity of the helicopter wake is less than the velocity of the helicopter), it will not obscure the boat wake to the front and sides of the helicopter. The

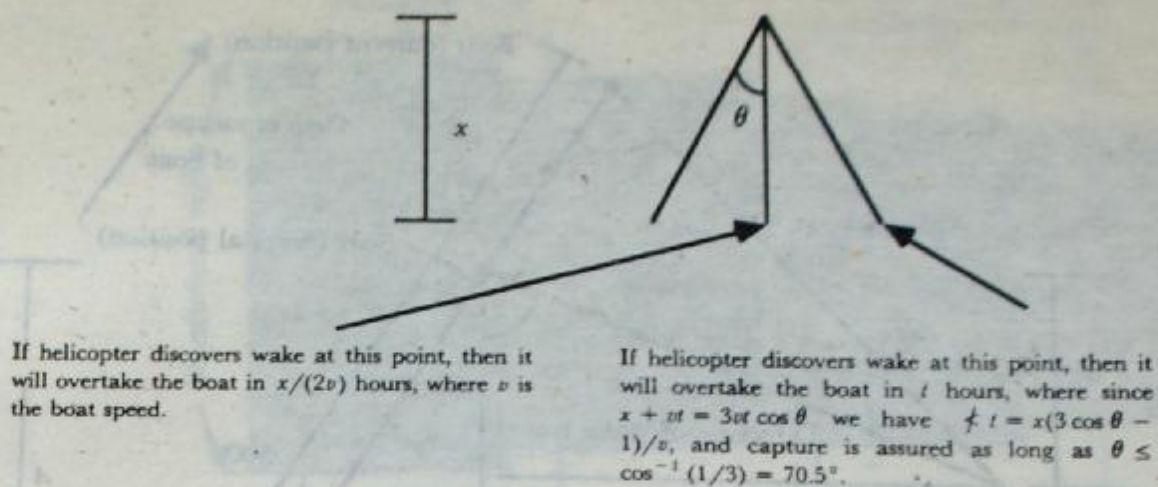


Figure 11. Capture time.

helicopter wake expands and can interrupt the water under the helicopter whenever the latter passes near its original path. However, this happening will not obscure the larger wake produced by the boat, although it may obscure the relatively undisturbed water between the major wakes produced by the boat. Therefore, we will neglect the helicopter wake.

The only escape method we investigate is method B, for which Figure 12 shows the critical case with helicopter just barely detecting the wake of the fleeing boat. As noted earlier in our assumptions, the wake is at 19.5° to the boat's centerline. Let v be the speed of the boat and x the distance traveled by the helicopter. Then we have

$$\begin{aligned} A &= 500 \sin \theta, & L &= y \cos \theta, \\ x &= 3vt, & y &= (\tan 19.5^\circ)vt = 0.35vt, \\ x &= y \sin \theta + 500 \cos \theta, & \text{so} \\ 3vt &= 0.35vt \sin \theta + 500 \cos \theta, & \text{or} \\ vt(3 - 0.35 \sin \theta) &= 500 \cos \theta. \end{aligned} \quad (1)$$

We also find

$$200 = A - L = 500 \sin \theta - y \cos \theta = 500 \sin \theta - 0.35vt \cos \theta$$

so

$$0.35vt \cos \theta = 500 \sin \theta - 200 \quad (2)$$

Dividing (1) by (2) eliminates vt :

$$\frac{3 - 0.35 \sin \theta}{0.35 \cos \theta} = \frac{500 \cos \theta}{500 \sin \theta - 200}$$

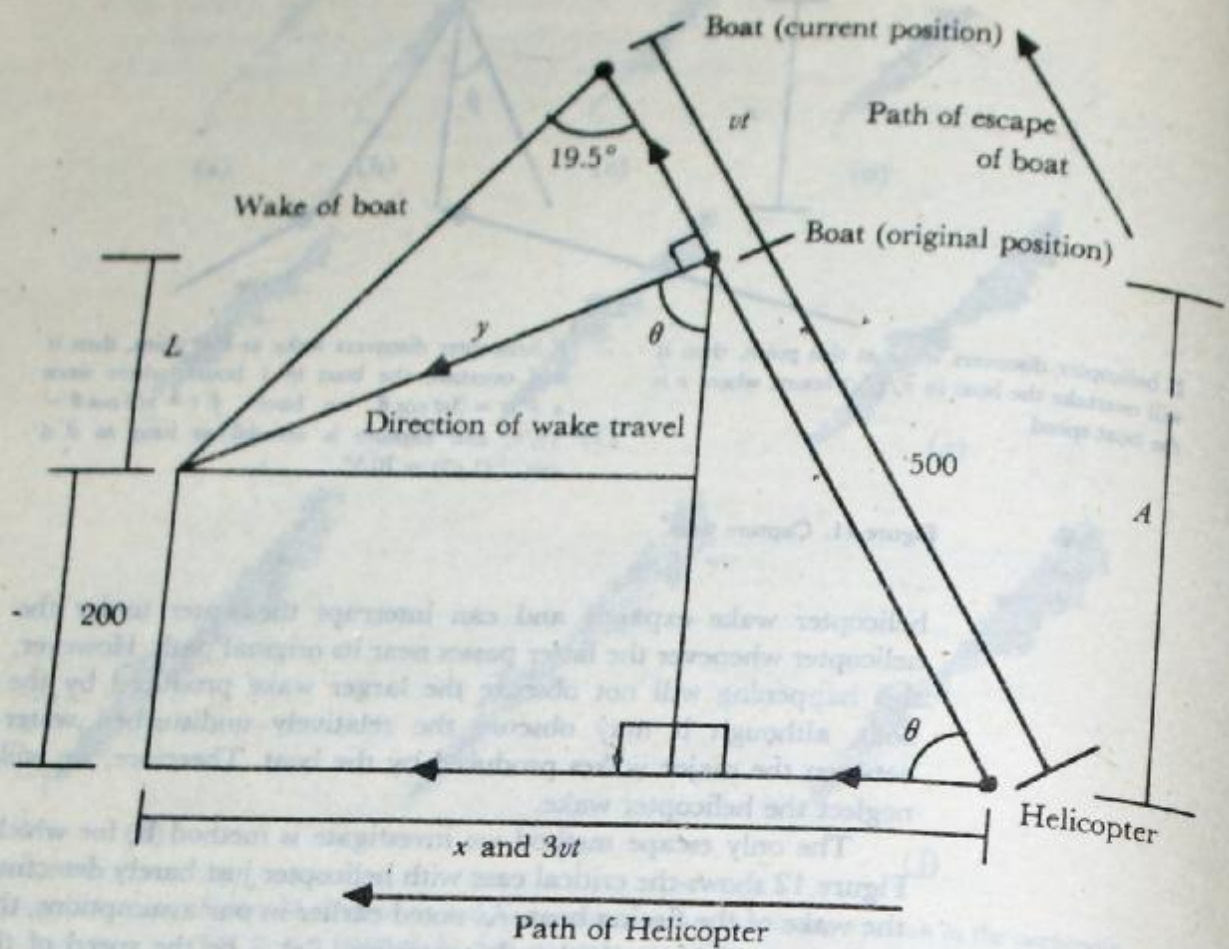


Figure 12. Finding escape distance with wake, escape method B (not drawn to scale).

Cross-multiplying yields

$$500(0.35)\cos^2 \theta = (500 \sin \theta - 200)(3 - 0.35 \sin \theta),$$

$$\cos^2 \theta = (500 \sin \theta - 200)(3 - 0.35 \sin \theta) / (500)(0.35),$$

$$\theta = \cos^{-1} \sqrt{(500 \sin \theta - 200)(3 - 0.35 \sin \theta) / (500)(0.35)},$$

Using Picard's method, we find

$$\theta = 33.7^\circ,$$

so the escape distance is $500 \sin \theta = 277.4$ and the capture path now has width 555 ft. The best path found is the zigzag path used primarily in the case of no wake and search method C. The path now crosses the region, turns around and moves 777 ft. along the region's major axis, and recrosses the region (see Figure 13). This

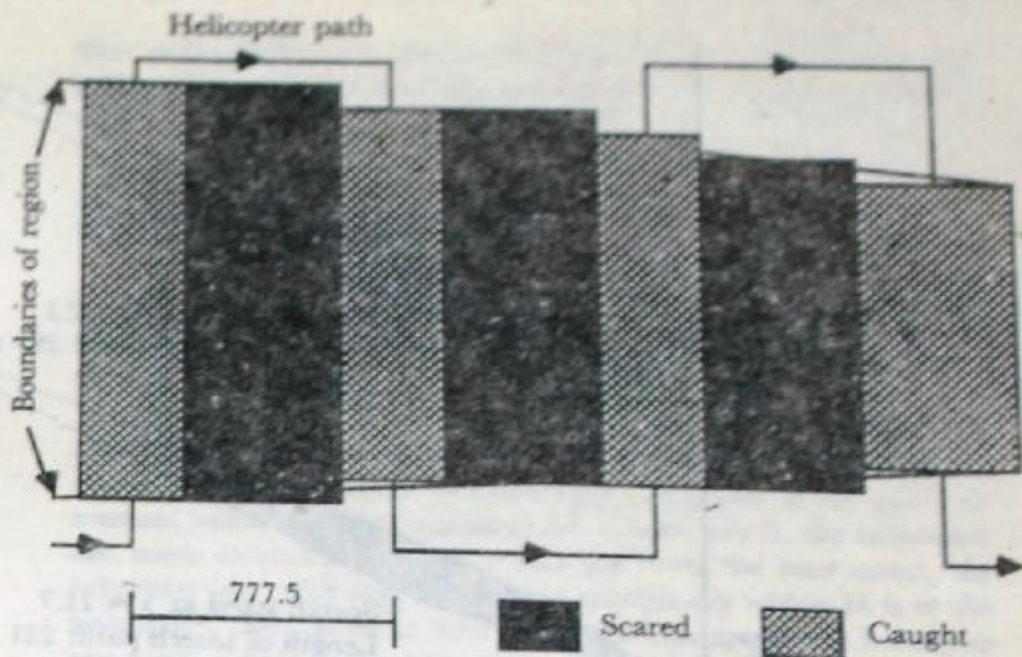


Figure 13. Details of final search method, over a small part of the search space.

way, a boat that moves as the helicopter passes on the first run is spotted from its wake on the second one. The farthest the boat can escape during this interval is $777 \text{ ft} + 2$ (maximum width of the region) which totals approximately 4.6 mi. The helicopter can catch the boat before it moves 1.6 mi by following the wake. The wake would not die down in the time the helicopter travels 4.6 mi (2 minutes).

There are a few parameters of the path that can be varied. One is the angle of crossing the region. Graphs at 20° , 60° , and 90° (measured from the centerline of the region, AC) show that there is little difference between them (see Figure 14). The only important differences are the exact shape of the region ignored when the helicopter turns toward base (so it doesn't run out of gas over water) and the number of turns. The exact shape of the region indicates that a 90° angle would be the best, since the less probability cut off, the higher the chance of finding the boat. The number of turns indicates that a smaller angle would be better, as the pilot is less likely to make mistakes if fewer turns must be made.

Since we assume that the helicopter has a range of 287.5 mi, and it must reach the search area and return, we can assume it can go a maximum of 225 mi in the search pattern. Subject to this constraint, we can try to improve our starting position. The approximate length of the region covered is 19 mi. Maximizing the probability in a 19 mi

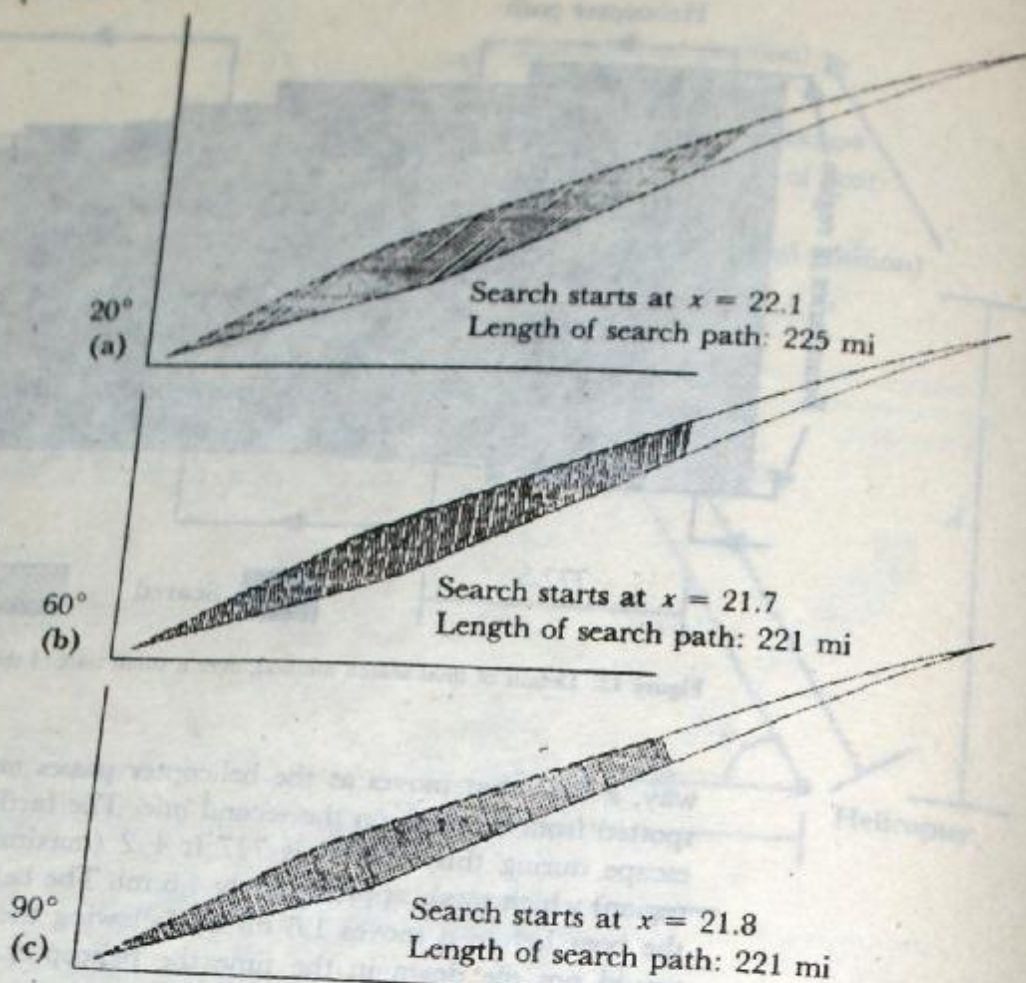


Figure 14. Search paths crossing the region at 20° , 60° , and 90° .

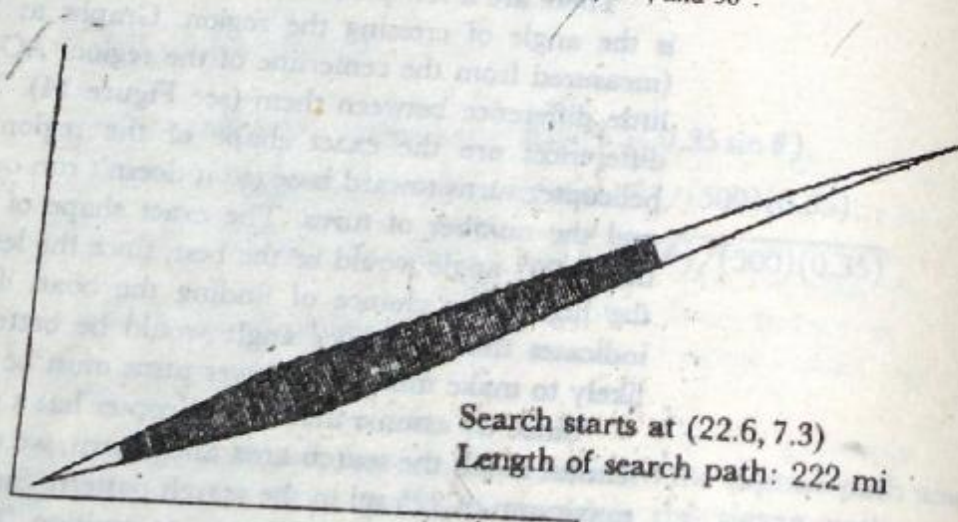


Figure 15. The optimal search path, for the case of the boat having a wake and pursuing escape method B.

slice yields starting coordinates of (22.6, 7.3) as in Figure 15. The graph of this path covers approximately 91% of the probability region.

Sensitivity Analysis

There are three separate sensitivity analyses in the program. The first applies to the no-wake case and the second and third apply to the with-wake case.

For the case of no wake and escape method C, we tested the sensitivity of the capture angle to the ratio of boat speeds. The results appear in Table 1. If the helicopter goes as slowly as the curve of constant width problem implies ($1.57 \times$ boat speed), the helicopter can never catch the boat. If it goes 2.4 times the boat speed, the helicopter can catch only a boat that is originally within 16 ft of the helicopter's path. Even at three times the boat speed, the helicopter can catch a boat only within 60 ft of the path. This value is quite sensitive, as the approximate derivatives of capture distance with respect to speed ratio are about 50, far too high to allow for possibly major differences between the boat's and helicopter's capabilities.

The second analysis, for the with-wake case and escape method B, tested sensitivity of the capture angle and distance to the four variables: wake angle, distance to hear the helicopter, ratio of speeds, radius of the searchlight. As Tables 2-4 show, none of these factors significantly affect the capture angle or distance, as the approximate

Table 1
Sensitivity of capture angle to the ratio of boat speeds
(case of boat with wake).

Ratio	d (cap. dist.)/ d (speed r.)	capture distance
1.57	103.4	-88.3
2.0	84.2	-24.7
2.2	77.2	-2.37
2.4	71.2	16.8
2.6	66.0	33.1
2.8	61.5	47.2
3.0	48.0	59.6
3.2	53.9	70.3
3.4	50.8	79.8
3.6	48.0	88.3
3.8	45.4	95.9

Table 2
Sensitivity of capture angle to wake angle (case of boat with wake).

Wake angle	$d(\text{cap. angle}) / d(\text{wake ang.})$	$d(\text{cap. dist.}) / d(\text{wake ang.})$	capture angle
18.5	.398	2.39	33.3
18.6	.398	2.90	33.4
18.7	.398	2.90	33.4
18.8	.399	2.90	33.4
18.9	.399	2.90	33.5
19.0	.399	2.90	33.5
19.1	.400	2.90	33.6
19.2	.400	2.91	33.6
19.3	.400	2.91	33.6
19.4	.401	2.91	33.7
19.5	0.000	0.000	33.7
19.6	.401	2.91	33.8
19.7	.402	2.92	33.8
19.8	.402	2.92	33.8
19.9	.402	2.92	33.9
20.0	.403	2.92	33.9
20.1	.403	2.92	34.0
20.2	.404	2.92	34.0
20.3	.404	2.93	34.0
20.4	.404	2.93	34.1

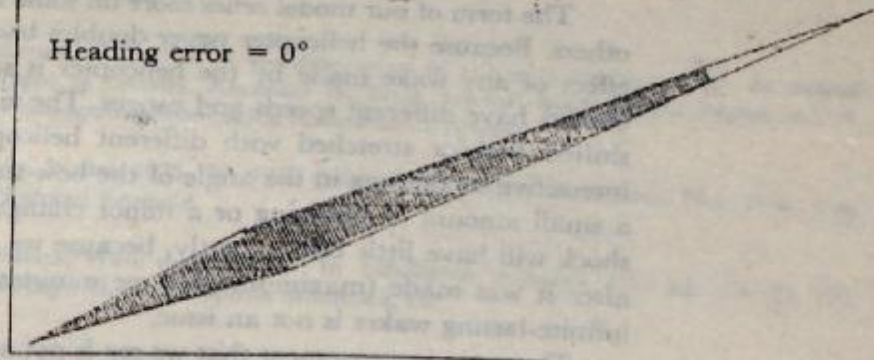
Table 3
Sensitivity of capture angle to distance at which the helicopter is heard (case of boat with wake).

Hear dist.	$d(\text{cap. ang.}) / d(\text{hear d.})$	$d(\text{cap. dist.}) / d(\text{hear d.})$	capture angle
450	-.0652	.137	36.98
460	-.0636	.1367	36.26
470	-.0621	.1362	35.58
480	.0606	.1357	34.93
490	.0592	.1353	34.31
500	0.0000	0.0000	33.72
510	.0566	.1345	33.15
520	.0554	.1342	32.61
530	.0542	.1338	32.09
540	.0531	.1335	31.59
550	.0521	.133	31.11

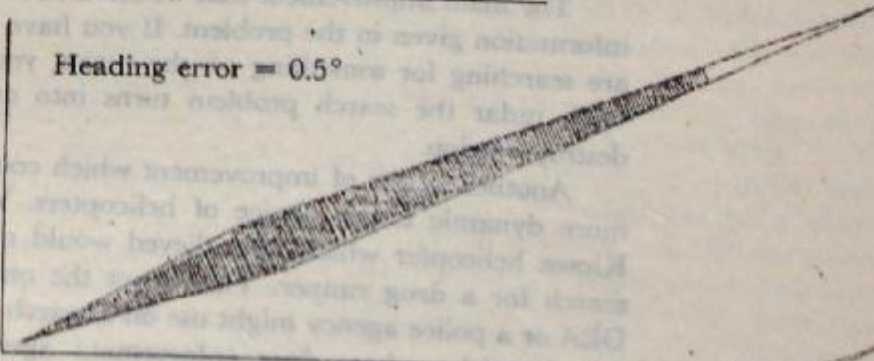
Table 4
Sensitivity of capture angle to speed ratio (case of boat with wake).

Speed ratio	$d(\text{cap. ang.})/$ $d(\text{speed r.})$	$d(\text{cap. dist.})/$ $d(\text{speed r.})$	capture angle
2.0	-3.7	-26.3	37.4
2.2	-3.3	-23.9	36.4
2.4	-3.0	-21.9	35.5
2.6	-2.8	-20.2	34.8
2.8	-2.6	-18.7	34.2
3.0	0.00	0.0	33.7
3.2	-2.3	-16.4	33.3
3.4	-2.1	-15.4	32.9
3.6	-2.0	-14.6	32.5
3.8	-1.9	-13.8	32.2

Heading error = 0°



Heading error = 0.5°



Heading error = 1°

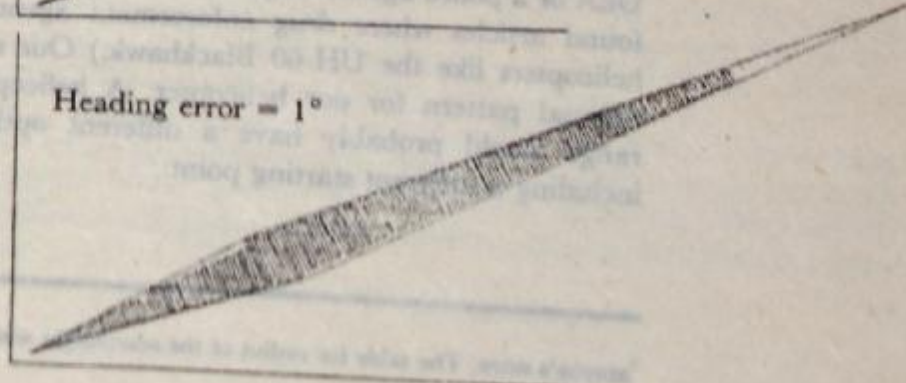


Figure 16. Sensitivity of solution to error in helicopter heading (case of no boat wake).

derivatives are quite small, except the ratio of the speeds.³ For the ratio of the speeds, the derivatives are -2.45 for the angle and -17.2 for the distance. This is a noticeable difference and a change in the speed ratio would drastically change any solution.

The third case is the sensitivity of the final solution, for the with-wake case and escape method B, to helicopter heading. The helicopter would have to be off by at least 0.5° to affect the search path significantly (see Figure 16). Since the problem says that the helicopter can follow the original 110° path perfectly, it seems safe to assume that the helicopter can keep its heading more accurate than 0.5° .

Improvement

The form of our model relies more on some assumptions than on others. Because the helicopter never doubles back over its path, the effect of any wake made by the helicopter is small. Different helicopters have different speeds and ranges. The zigzag pattern will be shifted and/or stretched with different helicopters. The model is insensitive to changes in the angle of the bow shock; this means that a small amount of planning or a minor change in the angle of the shock will have little effect. Lastly, because we cross any wake soon after it was made (maximum of three minutes), the assumption of infinite-lasting wakes is not an issue.

The main improvement that we see is not actually related to the information given in the problem. If you have a helicopter and you are searching for something on the ocean, you should have radar. With radar the search problem turns into an easy home-in-and-destroy mission.

Another source of improvement which could make our solution more dynamic is our choice of helicopters. We chose the OH-58 Kiowa helicopter which we believed would most likely be used to search for a drug runner. This is not the only helicopter that the DEA or a police agency might use on a search for a drug runner (we found articles where drug enforcement agents used sophisticated helicopters like the UH-60 Blackhawk.) Our search pattern is the optimal pattern for our helicopter. A helicopter with a different range would probably have a different optimal search pattern, including a different starting point.

³ EDITOR'S NOTE. The table for radius of the searchlight was not submitted.

The most serious problem is with our search result. We know how to find the boat. Once we find it, though, what do we do with it? We have data that show how far we can chase a boat and still return to our base. If we chase a boat beyond this limit, we can no longer return to base and we end up ditching into the sea. Even if we have enough fuel to return once we overtake the boat, are we supposed to ask its crew to stop so we can come aboard? The simple fact is that there is no practical way to deal with the boat with only a helicopter. In most open ocean drug busts, it is usually a helicopter that finds the drug boat and another ship that pulls up and stops the drug boat. The problem made no provisions for a boat. If it had, we could have found a better search pattern by tricking the drug boat into moving towards either the helicopter or the search boat.



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